

Lecture (18): The Optical System of the Eye

(Dr. Wagdy Nashaat Habeeb)

1. Optics of Vision

➤ Electromagnetic spectrum:

(A) Visible portion:

- Light
- Wavelength: 4000 – 7000 Å°
- Spectral colors: violet ← indigo ← blue ← green ← yellow ← orange ← red

(B) Invisible portion:

	1- <u>Ultraviolet rays:</u>	2- <u>Infrared rays:</u>
Wavelength:	< 4000 Å°	> 7000 Å°
Effects:	*Skin darkening *Vit. D3 activation *Bacterial killing	Heating effect

➤ Behavior of light in nature:

- Light velocity (v): in air (300.000 Km/s) > liquids > transparent solids
- When light rays fall on an object, they are either absorbed, reflected, or pass (with or without refraction)

- The degree of refraction depends on:

(1) Refractive index (RI) of medium: $RI = \frac{v \text{ of light in air}}{v \text{ of light in medium}}$

(2) Angle of incidence:

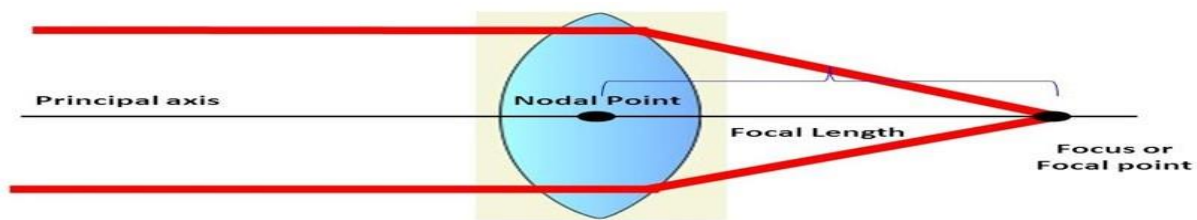
(a) Perpendicular rays → no refraction

(b) Forming an angle → refracted towards the perpendicular line if passed

from air to a denser medium & *vice versa*

➤ Lenses:

- Convex lenses → converge // rays
- Concave lenses → diverge // rays



- Lens refractive power (RP) “diopters” $\propto \frac{1}{\text{Focal length "meters"}}$
- Lens formula: $RP = \frac{1}{\text{Object distance}} + \frac{1}{\text{Image distance}}$

➤ In the eye:

(A) Application of the lens formula:

In the normal resting (non-accommodating) eye, light rays from an object placed at the infinity (a distance ≥ 6 m) are focused on the retina (15 mm behind the nodal point of the lens), so:

$$RP \text{ of the resting eye} = \frac{1}{\infty} + \frac{1}{0.015} = 0 + 67 = 67 \text{ diopters}$$

(B) Size of retinal image:

It can be calculated if the size of the object & its distance from the eye are known by the following formula:

$$\frac{\text{Size of object}}{\text{Size of image}} = \frac{\text{Distance of object from lens (in meters)}}{\text{Distance of image from lens (0.015 meters)}}$$

(C) Refractive media of the eye:

Refractive medium	RI	RP
1. Cornea:	1.38	45 diopters
2. AH:	1.33	
3. Lens:	1.4	20 diopters
4. VH:	1.34	

N.B. The greatest light refraction occurs at the cornea (Why?)

2. Accommodation

➤ Definition:

Ability of the lens to ↑ its RP (mainly *via* ↑ the curvature of its ant. surface) to focus the image of near objects on the retina

➤ Mechanism:

During near vision → Ciliary ms. contracts → SL become lax → Lens becomes more spherical

➤ Proves that accommodation affects mainly the ant. surface:

1. ↓↓ depth of ant. chamber (slit lamp)
2. Purkinje-Sanson images

➤ Power of accommodation:

= RP of lens with maximal accommodation (during near vision) – RP of lens of resting eye (during far vision)

“It ↓↓ with age”: 14 D at 10 years (*i.e.*, RP can increase from 20 D to 34 D), 10 D at 20 years, 2 D at 50 years, 1 D at 60 years & 0 at 70 years

➤ Range of accommodation:

= distance between far point & near point

“It ↓↓ with age”

Far point: The farthest point from eye which can be seen clearly without accommodation (6 m in emmetropes)

Near point: The nearest point from eye which can be seen clearly with maximal accommodation (10 cm in young adults & 80 cm at 60 years)

➤ Presbyopia:

*Definition: ↓↓ power of accommodation by aging

*Causes: 1. ↓↓ elasticity of lens capsule 2. Hardening of lens (due to protein denaturation)

*Effects: 1. Near point: recedes away from eye
2. Far point: does not change
3. Range of accommodation: ↓↓

*ttt: Convex lens during reading & near vision (or) Bifocal glasses: with its upper segment normally focused for far vision & its lower segment focused for near vision

➤ Errors of refraction:

Emmetrope eye:

Normal eye in which:

1. // rays from far objects are focused on retina when accommodation is completely relaxed
2. near objects are focused on retina with accommodation

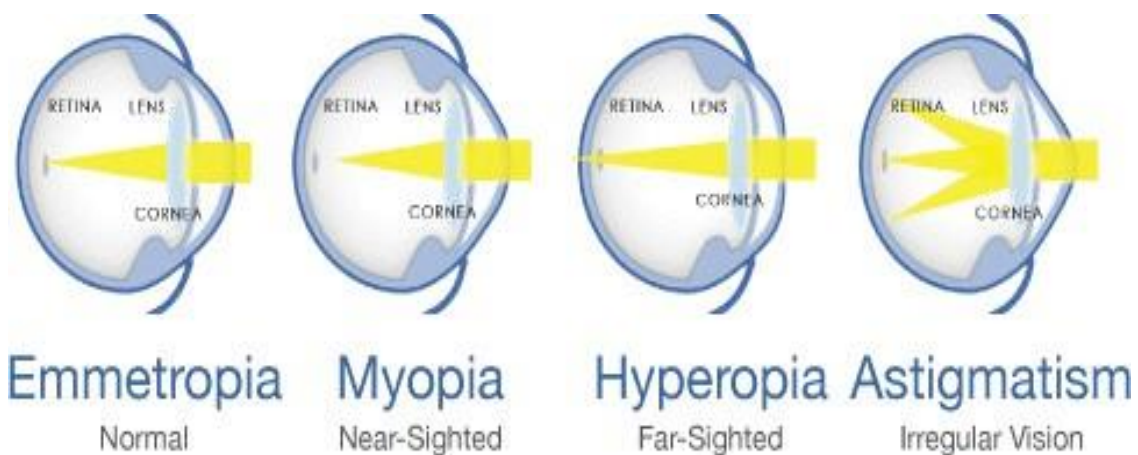
Ametropia:

Errors of refraction.

It includes: 1. Myopia 2. Hypermetropia (Hyperopia) 3. Astigmatism

Anisometropia:

Unequal RP in both eyes → treated by corrective lenses with different appropriate RP for each eye



	1. MYOPIA “Short-sightedness”	2. HYPERMETROPIA “Far-sightedness”
<u>*Definition:</u>	Eye defect in which // rays are focused in front of retina	Eye defect in which // rays are focused behind retina

* <u>Causes:</u>		
(a) Eyeball:	Abnormally long	Abnormally short
(b) RP of eye:	↑↑ (due to ↑↑ curvature of lens or cornea)	↓↓ (due to ↓↓ curvature of cornea mainly)
* <u>Effects:</u>		
(a) Clinically:	*<u>Far vision:</u> Patient can't see far objects (no mechanism to ↓↓ RP of lens) *<u>Near vision:</u> Not impaired	*<u>Far vision:</u> Patient uses accommodation for far vision → continuous ciliary ms. cont. → eye discomfort & headache

(b) Near point:		* <u>Near vision:</u> Near objects must be placed farther from eyes to be seen clearly
(c) Far point:	< 10 cm	> 10 cm
(d) Accommodation:	< 6 m	(?)
	Aggravates the condition	Essential for near & far vision
* <u>ttt:</u>	Concave (divergent) lenses	Convex (convergent) lenses

3. Astigmatism:

*Definition: An error of refraction in which light rays are not focused to a single point on the retina

*Cause: Irregular curvature of the cornea, or less commonly the lens, in different meridians (planes). In most cases, the cornea has maximum curvature in the vertical meridian & least curvature in the horizontal meridian → called “*astigmatism with the rule*”. When reverse conditions are present → called “*astigmatism against the rule*”

*Effects:

- (a) Blurred or distorted vision
- (b) A point is seen as a line & a line appears to have a halo on either side
- (c) Astigmatism may be myopic, hyperopic or both

*ttt: Cylindrical lens, with its curvature // to the plane to be corrected (?)

Lecture (19): The Visual Process

(Dr. Wagdy Nashaat Habeeb)

1. Duplicity Theory of Vision

“2 types of photoreceptors (rods & cones) → 2 separate mechanisms of vision (scotopic & photopic)”

	Rods	Cones
(A) <u>Anatomical:</u>		
1. Number:	120 million in each retina	6 million in each retina
2. Site:	Peripheral (mainly)	Central (mainly) “Fovea centralis contains cones only”
3. Photopigment:	*More *1 type (rhodopsin)	*Less *3 types (blue, green & red)
4. Convergence:	200:1 *200 rods → 1 bipolar cell *Many bipolar cells → 1 ganglion cell	1:1 (no convergence) *1 cone → 1 bipolar cell *1 bipolar cells → 1 ganglion cell
(B) <u>Physiological:</u>		
1. Light sensitivity:	Very high (Up to 300 times as cones)	Very low
2. Visual acuity:	Very low	Very high
3. Color perception:	No	Very high
4. Vision:	Scotopic (dim / night) vision	Photopic (bright / day) vision